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A Bearing Fault Diagnosis Method Based on Enhanced Singular Value Decomposition

Hua Li, *Student Member, IEEE*, Tao Liu, Xing Wu, and Qing Chen

***Abstract*—For the two shortcomings of singular value decomposition (SVD), the determination of the reconstruction order and the poor noise reduction ability, an enhanced SVD is introduced in this article. The core ideas include: first, an efficient method to determine the reconstructed order of SVD and the relative-change rate of the singular envelope kurtosis is presented, composed of improved SVD (ISVD). Then, the method to select the optimal node of wavelet packet transform (WPT) by the criterion of envelope kurtosis maximum is presented, composed of improved WPT (IWPT). The flexible filter design and superior noise reduction abilities of the IWPT and the passband denoise ability of the ISVD are organicly combined to form enhanced singular value decomposition (E-SVD) method. In addition, an indicator is introduced to evaluate the performance of the results. First, the reconstructed signal is obtained by performing ISVD on the original signal. Second, IWPT is executed on the reconstructed signal to achieve the optimal node. Finally, the filtered signal is combined with the envelope power spectrum to extract the bearing fault characteristic frequency. The method's validity and superiority are verified by the analysis of simulated data and actual cases of rolling bearing.**

***Index Terms*—Envelope kurtosis, fault diagnosis, reconstructed order, rolling bearing, singular value decomposition (SVD), wavelet packet transform (WPT).**

I. Introduction

ROTATING machinery plays a pivotal role in our modern lives, which is widely used in a variety of large, medium, and small mechanical equipment and various power electronics equipment, such as electric machines, wind turbines, trains, and power generators. Rolling bearing is one of the most critical components in rotating machinery [1]–[3]. Its failure often leads to the mechanical equipment breakdown, which is the main cause of mechanical equipment failure. Therefore, research on the fault diagnosis of rolling bearing in rotating equipment is finally attracting the attention of researchers [4]–[6].

Extraction of fault features is the key to achieve fault diagnosis of bearing. When the different parts of the bearing fail, then different characteristic frequencies will appear. Studies have found that through amplitude demodulation analysis, the envelope signal can be achieved and it has more fault-related characteristic information than the original signal [7]. In the initial stage of bearing failure, the fault signal is very weak. It inevitably contains a large amount of low-frequency noise when acquiring the vibration signal, which will seriously influence the results of the envelope spectrum analysis. If there is local failure in the rolling bearing, the vibration signal has a distinctly nonstationary nature. Therefore, how to realize fault diagnosis of bearing has always been a key issue in the field of fault diagnosis of rotating equipment. Naturally, various relevant research results have been carried out [8]–[10].

Signal processing method is the foundation for implementing fault characteristics extraction and thus fault diagnosis of the various rotating equipment. Singular value decomposition (SVD) is one of the frequently used methods. Reconstructed order and noise reduction are essential to SVD research. Therefore, subsequent reviews will proceed from these two aspects. Throughout recent studies, many indicators and methods have been suggested to achieve effective selection of reconstructed order [11]–[15]. Although the selection method [15] gave a quantitative indicator, but it is based directly on singular values as shown in [14]. On the contrary, because of the poor noise reduction capability of SVD, scholars have made significant efforts to enhance its noise reduction ability to achieve gratifying results. They have focused on SVD research works in combination with other methods, such as Hilbert–Huang transform [16], singular spectrum analysis [17], local mean decomposition [18], empirical modal decomposition (EMD) [19], and the method wavelet transform (WT), locally linear embedding [20], and sparse representation [21]. There are plentiful examples of methods, just to name a few.

Next, a brief overview of wavelet packet transform (WPT) is given here. To illustrate WPT, we first describe the WT briefly. WT has superior local property in both time and frequency domains [22]. However, the WT have deficiencies: When WT is used, the wavelet function derived using a basis function

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is difficult to accurately approximate the local signal characteristics at different scales. WPT can overcome the defects effectively. Therefore, WPT has more extensive applications in the field of fault diagnosis [23]–[25]. In view of the superior noise reduction capability of WPT filter, this article combines improved SVD (ISVD) and improved WPT (IWPT) to realize the complementary advantages of the two methods and forms a new and effective fault diagnosis method.

In this article, the reconstructed order determination and the noise reduction ability enhancement of SVD are both studied to put forward enhanced SVD (E-SVD). First, for the purpose of determining the model reconstruction order of SVD efficaciously, a novel quantitative indicator singular envelope kurtosis (SEK) is introduced at first, and the ISVD is formed. Second, the method of selecting the optimal node of WPT using the criterion of envelope kurtosis maximum (CEKM) indicator is presented, and the IWPT is formed. Third, the proposed method fully combines the flexible filter design and superior noise reduction abilities of the IWPT with the passband denoise ability of the ISVD. It greatly enhances the ISVD's ability to remove noise to achieve the fault diagnosis accurately. Therefore, the article introduces a novel and effective bearing fault diagnosis method named E-SVD. Finally, to evaluate the presented E-SVD method better, the δ efficacious indicator is proposed. In addition, it has been compared with the methods in [14], [19], and [23], to provide convincing evidence for its more robust performance. The validity and superiority of the proposed E-SVD method are verified by results of the analysis of simulated signal and experimental cases.

The rest of this article is organized as follows. Section II-A describes the ISVD based on the original selection method SEK, Section II-B presents the IWPT based on the node selection method CEKM, Section II-C gives the indicator δ, and Section II-D elaborates on the fault diagnosis method E-SVD. In Sections III and IV, the introduced E-SVD method is verified by a simulated signal and the actual cases of rolling bearing. The comparisons of the presented E-SVD method with [14], [19], and [23] are conducted. Finally, Section V concludes this article.

II. Proposed Method

In this article, an E-SVD method is proposed for the problems that need to be studied in the original SVD method.

A. ISVD Method

The SVD theory has been elaborated in various papers, such as [16]–[19]. For the sake of space, it will not be described here. Interested readers can refer to [26].

The kurtosis has the characteristic of being sensitive to the impact signal, whereas the signal of local failure of rolling bearing always exhibits impact features obviously, and it is widely used [27]. Although the kurtosis based on the time signal is effective under some conditions, its performance is poor in the low signal-to-noise ratio (SNR) state because the potential periodic peak is submerged by severe undesired interference. In addition, non-Gaussian noise may lead to a larger kurtosis value. Thus, an indicator SEK is introduced to calculate the kurtosis from the Hilbert transform (HT) and select the effective reconstructed order of SVD.

Here, the original discrete signal is $X = [x(1), x(2), \ldots, x(N)]$, performing SVD on the signal. Then, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 0$ can be obtained. We assume that the reconstructed order is i $(i = 1, 2, \ldots, k)$. The steps are given as follows: First, the envelope of the first i-order reconstructed signal is obtained; then, the envelope kurtosis of this first i-order is obtained; finally, the envelope kurtosis value of each reconstructed order i is compared to obtain the optimal reconstructed order whose envelope kurtosis value is maximum.

Envelope the reconstructed signal of the first i-order, i.e.,

$$xt_r^i = \left| \frac{1}{\pi t} \cdot x_r^i(t) \right|, \quad i = 1, 2, \ldots, k \tag{1}$$

where $x_r^i(t)$ is the reconstructed signal of the first i-order and xt_r^i is the absolute value (the HT result of the reconstructed signal).

Naturally, the envelope kurtosis value of the reconstructed signal by the first i-order can be calculated as

$$dek_r(i) = \frac{E\left(xt_r^i - \mu(xt_r^i)\right)^4}{\sigma(xt_r^i)^4}, \quad i = 1, 2, \ldots, k \tag{2}$$

where $\mu(xt_r^i)$ denotes the mean of xt_r^i and $\sigma(xt_r^i)$ denotes the standard deviation of xt_r^i.

The concept of SEK can be defined as follows:

$$\mathrm{SEK}_i = \frac{dek_r(i) - dek_r(i+1)}{dek_r(i+1)} \quad (i = 1, 2, \ldots, k-1) \tag{3}$$

where $dek_r(i)$ represents the envelope kurtosis value of the reconstructed signal with first i-order and k denotes the number of singular values. Then, calculating SEK_i under each order i, and SEK_i with the maximum absolute value can be obtained. It is assumed that the maximum of the absolute value of SEK_i is obtained in the reconstructed signal of the first i-order, which can be recorded as

$$\mathrm{SEK}_{i,\max} = \left| \frac{dek_r(i) - dek_r(i+1)}{dek_r(i+1)} \right| \quad (i = 1, 2, \ldots, k-1). \tag{4}$$

When the value of the absolute maximum $\mathrm{SEK}_{i,\max}$ is obtained (whether it comes from a positive or negative value SEK_i), it can be considered that when SEK_i is the absolute maximum $\mathrm{SEK}_{i,\max}$, the optimal order of the SVD model can be also determined.

In order to describe the SEK for selecting the reconstructed order more clearly, it is necessary to discuss the following two cases in detail. The following formula is shown

$$\mathrm{SEK}_{i,\max} = \begin{cases} \frac{dek_r(i) - dek_r(i+1)}{dek_r(i+1)} & (5) \\ -\frac{dek_r(i) - dek_r(i+1)}{dek_r(i+1)} & (6) \end{cases} \quad (i = 1, 2, \ldots, k-1).$$

They are elaborated as follows: 1) If $dek_r(i) > dek_r(i+1)$, that is, $dek_r(i) - dek_r(i+1) > 0$. The $\mathrm{SEK}_{i,\max}$ is obtained by the positive value, which is shown in formula (5). It indicates that the largest sudden drop occurs at the ith order. Thus, the ith order is considered to be the optimal reconstructed order. 2) If $dek_r(i) < dek_r(i+1)$, i.e., $dek_r(i) - dek_r(i+1) < 0$. The

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$\mathrm{SEK}_{i,\max}$ is obtained by the negative value, which is shown in formula (6). It indicates that the envelope kurtosis value of the ith order is significantly smaller than the value of $i+1$th order, and the first $i+1$th order is chosen as the optimal order.

Here are some detailed discussions of the SEK method. From the detailed theoretical analysis above, we know that the SEK method is based on the envelope kurtosis of reconstructed signal of each order. In statistics, kurtosis is defined as the peak or flatness of the frequency distribution in [27]. SEK is utilized to obtain kurtosis from the frequency domain, which has the advantage of kurtosis sensitivity to shock signals, and effectively overcomes the shortcoming of kurtosis affected by strong interference and non-Gaussian noise. Therefore, the selection method is based on SEK starting from the impacts of the signal, using the sensitivity of the kurtosis to shock, based on the kurtosis theory. Additionally, it is better to determine the model order by the SEK method theoretically than directly determine the model order from the singular value size as shown in [14] and [15], and when the noise is intense, the results are often inaccurate. This is because when the SNR is low, the methods of [14] and [15] perform smoothly. The SEK can overcome the bad phenomenon effectually.

Based on the SEK indicator, the optimal reconstructed order of SVD can be effectively selected, i.e., the ISVD is achieved.

B. IWPT Method

The high frequency band of WT has poor frequency resolution. WPT can overcome the deficiency of WT to enhance the frequency resolution effectively. They can be implemented based on a series of wavelet filters [28], [29].

Following the analysis result above, the reconstructed signal can be generated. WPT is used to enhance the denoising ability of ISVD. WPT with certain depth is executed to obtain a series of nodes. Here, an effective node selection method based on CEKM, which is formally defined as follows.

1) Performing WPT on the reconstructed signal, a set of nodes are obtained. Envelope kurtosis values of each node at different depths are calculated.
2) Comparing the envelope kurtosis values to obtain the maximum. Then, this node is selected as the optimal node for subsequent analysis. The specific calculated process is given in detail as follows.

Performing WPT on the reconstructed signal $x_r(t)$. WPT's depth is $l\ (l = 1, 2, \ldots, L)$. There are $J + 1\ (J = 2^l - 1)$ nodes in the lth layer. The envelope kurtosis value is obtained for each node of each depth, which is marked as $dek_{l,j}$ (l denotes the depth of decomposition, and j denotes the jth node of the lth layer). Therefore, $J + 1$ envelope kurtosis values can be obtained at the lth layer

$$\mathbf{DEK_l} = (dek_{l,0}, dek_{l,1}, \ldots, dek_{l,J}), \quad j = 0, 1, \ldots, J. \tag{7}$$

From (7), we can obtain the envelope kurtosis value of each node in the depth $l\ (l = 1, 2, \ldots, L)$ and the maximum value in the lth depth. Thus, L maximum values can be obtained as shown below:

$$\mathbf{DEK_{max}} = (dek^1_{\max}, dek^2_{\max}, \ldots, dek^L_{\max}) \tag{8}$$

where $dek^l_{\max}\ (l = 1, 2, \ldots, L)$ denotes the envelope kurtosis maximum of the nodes of each layer. The optimal node $(k^*(l, j))$ is located $(\mathrm{DEK}^l_{\max} = \max(\mathbf{DEK_{max}}))$, i.e.,

$$k^*(l, j) = \arg\ \max(\mathbf{DEK_{max}}). \tag{9}$$

The analysis in this section constitutes the IWPT method, which is utilized to process the reconstructed signal of ISVD to enhance the noise reduction capability.

C. Extraction of Fault Feature Frequency

Here, the envelope power spectrum analysis is executed on the reconstructed signal (based on the optimal node $k^*(l, j)$) to extract fault signature frequency.

In addition, the index δ (ratio of the sum of the first three harmonic amplitudes of the fault characteristic frequency to the maximum amplitude of the undesired interference frequencies.) is introduced as a supporting evidence to evaluate the performance of the noise removal capability of the E-SVD method

$$\delta = \frac{A\left(f_\bullet\right) + A\left(2 \cdot f_\bullet\right) + A\left(3 \cdot f_\bullet\right)}{\max\left(A(f_{\mathrm{Interference}})\right)}. \tag{10}$$

where $f_\bullet$ is the fault signature frequency, which can be derived from any of the four fault forms of the bearing. $A(f_\bullet)$, $A(2 \cdot f_\bullet)$, and $A(3 \cdot f_\bullet)$ represent the amplitude of the fault characteristic frequency, its second harmonic, and third harmonic, respectively. $f_{\mathrm{Interference}}$ are the interference frequencies, $A(f_{\mathrm{Interference}})$ are the amplitudes of the interference frequencies, $\max(A(f_{\mathrm{Interference}}))$ represents the amplitude maximum of the interference frequencies.

The larger the indicator δ, the greater the proportion of the fault characteristic frequency in the spectrum obtained. That is, the larger the indicator δ, the better the effect.

D. Proposed E-SVD Method

Here, we introduce E-SVD for fault diagnosis of rotating machinery. Our goals are selecting the optimal reconstructed order and enhancing the noise reduction ability of SVD. Therefore, ISVD and IWPT are proposed separately. Combining IWPT's abilities of flexible filter design and superior noise reduction with the ISVD's pass-band denoising ability to achieve fault diagnosis better.

The specific steps and flow chart are given below.

1) The ISVD is utilized. The optimal reconstructed order of SVD of the original signal is determined by the proposed SEK index to obtain the reconstructed signal.
2) The IWPT is performed on the reconstructed signal. WPT is performed on the reconstructed signal at different depths. Calculating the envelope kurtosis of each node at different depths. Then, the node where the envelope kurtosis maximum is located is obtained by comparison. As shown in Fig. 1, the node is selected as the optimal node $k^*(l, j)$.
3) The selected optimal node $k^*(l, j)$ is refactored to get the reconstructed signal. The HT is executed on the reconstructed signal to obtain the envelope, and then the power spectrum of the envelope is analyzed.

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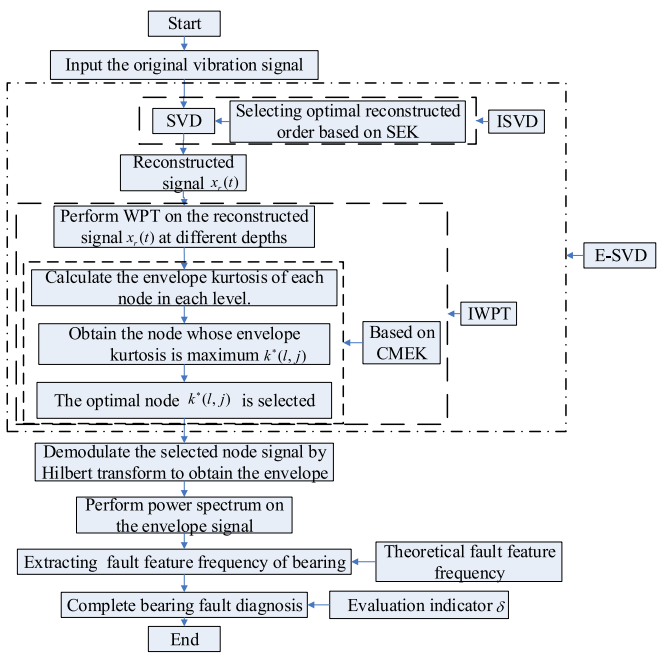

Fig. 1. Proposed method E-SVD.

4) The envelope power spectrum is utilized to pick up the fault feature frequency. Accurate fault identification can be achieved by comparison with the result obtained by theoretical calculations. In addition, the indicator δ is utilized as an auxiliary tool to evaluate the validity of the analysis results.

III. Simulated Analysis

For the purpose of simulating the failure of rotating equipment, the analog signal of rolling bearing with inner-race fault is used as an example, and the fault diagnosis method E-SVD is verified. The simulated signal can be obtained from [7]. Here, $f_s = 12\,000$ Hz, $f_n = 3000$ Hz, $f_i = 90$ Hz, $f_r = 17$ Hz, and the damping ratio $B = 500$. Gaussian white noise with an intensity of –10 dB is added to this simulated case.

The basic analytical results of the original signal are described in Fig. 2. Severe interference noise drowns the pulse of the analog signal, resulting in an unsatisfactory analysis result. Thus, the E-SVD is used to process the original vibration signal.

A. Analysis of ISVD Based on SEK

ISVD is executed on the simulated signal so that a series of SEK_i values can be obtained. Since the SEK_i has been flattened at a higher order and in order to be able to clearly display the SEK_i, the first 100 points of the relationship between the envelope kurtosis and the order are plotted. From Fig. 3, the largest sudden drop occurs at the second SEK value [absolute maximum is come from formula (5)]. Thus, the second order is selected for reconstruction and a noise-reduced signal is obtained.

Envelope analysis is applied on the above result. The rotational frequency f_r, the fault feature frequency f_i, and the

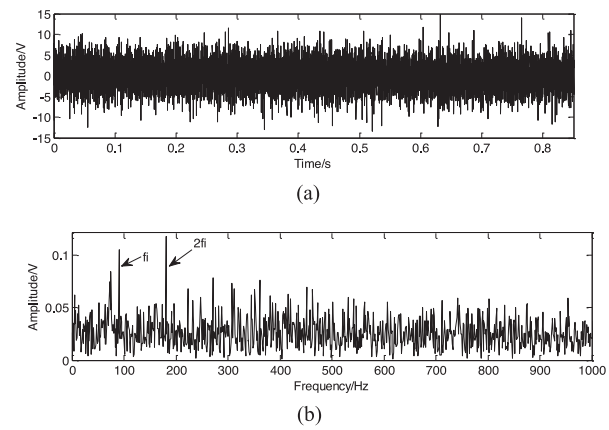

Fig. 2. Simulated signal with inner-race fault. (a) Vibration signal in time domain. (b) Result of the envelope spectrum analysis.

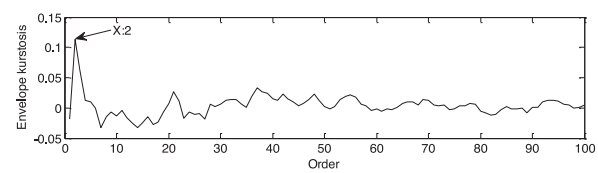

Fig. 3. SEK_i.

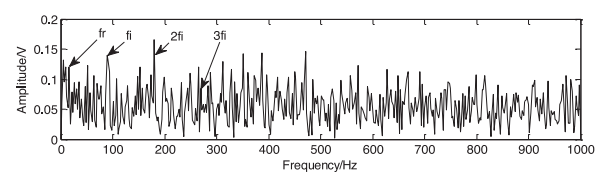

Fig. 4. Envelope spectrum of reconstructed signal.

harmonics ($2f_i$, $3f_i$) can be extracted from Fig. 4, and when it is compared with Fig. 2(b), the noise on the pass-band is effectively eliminated (It obviously becomes sparse throughout the frequency band.). However, its performance is not ideal. The IWPT is executed on the reconstructed signal to enhance the noise reduction capability of ISVD.

B. IWPT Analysis

The depth of IWPT is set as 4 in this article. Envelope Kurtosis values of all nodes at different depths i ($i = 1, 2, 3, 4$) are calculated. We compare the envelope kurtosis values to obtain the maximum. An envelope kurtosis maximum is available at each depth, respectively, $k(1,1)(4.3565)$, $k(2,3)(4.8580)$, $k(3,2)(5.5416)$, and $k(4,4)(7.6016)$. We compare them to obtain the global maximum, which is $k(4,4)$. Thus, the optimal node $k^*(4,4)$ can be obtained. Fig. 5 shows the envelope kurtosis values at different depths.

The reconstructed signal can be obtained from $k^*(4,4)$. Then, the power spectrum of the envelope is obtained, which is plotted in Fig. 6. In Fig. 6, the rotational frequency f_r, the fault feature

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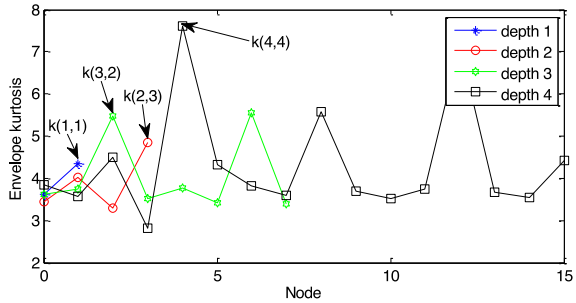

Fig. 5. Envelope kurtosis of different nodes at different depths.

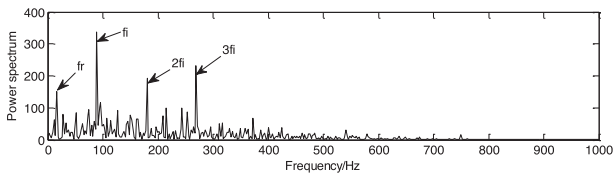

Fig. 6. Power spectrum of the envelope of the optimal node $k^*(4,4)$.

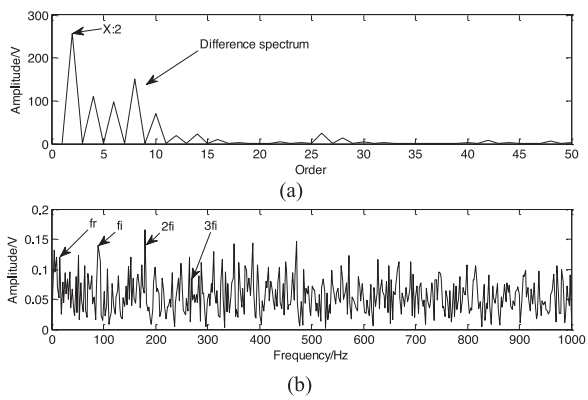

Fig. 7. Analysis result of [14]. (a) Difference spectrum. (b) Envelope spectrum of [14].

frequency f_i, and the harmonics $(2f_i, 3f_i, 4f_i)$ are very evident. The index $\delta = 6.61$ is calculated.

C. Comparisons With Other Methods

In order to provide convincing evidence that the introduced method has a more robust performance, it is compared with the methods in [14], [19], and [23].

First, the method in [14] is applied to analyze the same signal. Singular differential spectrum is directly used for fault diagnosis. The result is provided in Fig. 7. From Fig. 7(a), the reconstructed order is determined to be 2. It has the same reconstructed order as determined by the proposed method. In the envelope spectrum in Fig. 7(b), we can see that some of the passband noise is removed compared to Fig. 2(b), but the overall desiccation capability is weak and is interfered by strong noise. When it is compared with Fig. 6, the proposed method has considerable advantages, obviously. In addition, the indicator $\delta = 2.65$, which

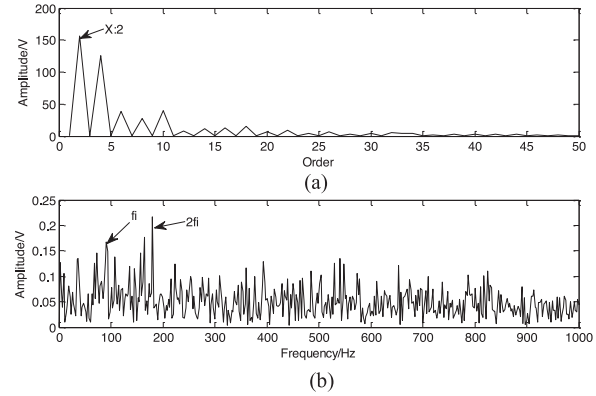

Fig. 8. Analysis results of [19]. (a) Difference spectrum. (b) The envelope spectrum.

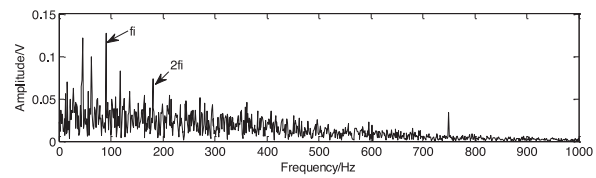

Fig. 9. Analysis result of [23].

is much smaller than the value of Fig. 6. This also illustrates the advantages of the E-SVD.

Second, the method in [19] is applied. EMD is used to improve the denoising ability of SVD. According to its idea, IMF1 is directly utilized, and the difference spectrum is executed on the IMF1 to acquire the model order. The results are plotted in Fig. 8. The reconstructed order is 2 depicted in Fig 8(a). Although the fault feature frequency f_i, and the $2f_i$ can be identified from Fig. 8(b). The entire frequency band is disturbed by noise. Obviously, Fig. 6 has more excellent performance. In addition, the indicator, $\delta = 2.9$ is much smaller than that of Fig. 6. This also provides evidence that E-SVD has considerable advantage.

Third, the method in [23] is used here. WPT is performed on the original signal and the kurtosis maximum is used to select the optimal node. After the analysis, it was found that the kurtosis value of node $k(4,12)(7.1095)$ is the largest; therefore, it is selected as an optimal node for envelope demodulation analysis, which is plotted in Fig. 9. Comparing it with Fig. 6, the proposed method has significantly better effect. The indicator $\delta = 2.07$ is calculated. Obviously, δ of the E-SVD is larger.

IV. Actual Signal Analysis

In order to provide sufficient evidence of the effectiveness and advantage of the E-SVD, it is applied to actual cases.

A. Case 1: Actual Bearing Signal From Case Western Reserve University (CWRU)

The case is an actual bearing with outer-race fault. It is taken from the public data provided by CWRU [30]. The data come from the fan end. Concrete parameters—bearing type:

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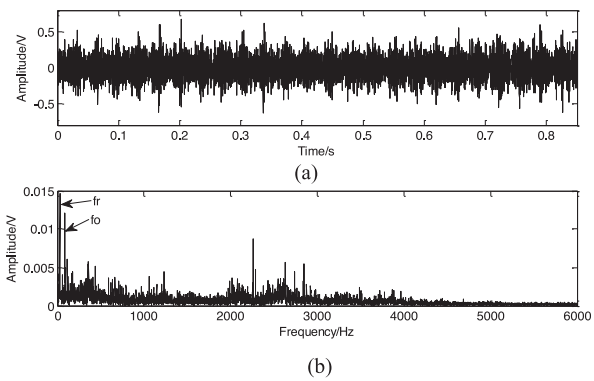

Fig. 10. Original signal analysis. (a) Time-domain waveform (outer-race fault). (b) Envelope spectrum (outer-race fault).

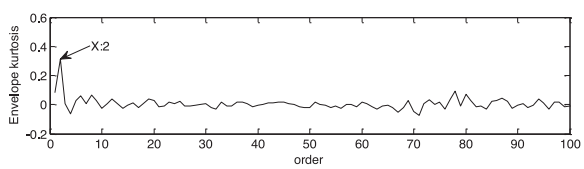

Fig. 11. SEK_i.

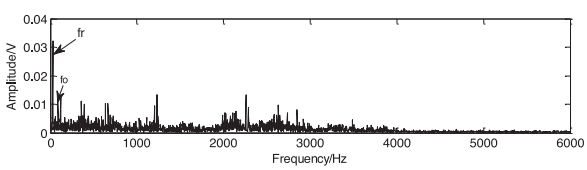

Fig. 12. Envelope spectrum analysis of the reconstructed signal.

SKF 6203-2RS, outer-race fault size: 0.021", spindle speed: 1728 r/min, sample frequency: $f_s = 12\,000$ Hz, and rotational frequency $f_r = 28.8$ Hz. The outer-race fault feature frequency is $f_o = 87.9$ Hz.

Analysis results are plotted in Fig. 10(a) and (b), respectively. Due to the severe noise, the impulse characteristics cannot be identified clearly. From Fig. 10(b), although we can extract the rotational frequency f_r and the fault feature frequency f_o, it is interfered by broadband noise of the entire frequency axis (There are large frequency peaks in the range of 2000 to 3000 Hz.). Therefore, the original signal is processed by the E-SVD.

1) ISVD Analysis Based on SEK: Here, ISVD is applied to the above signal. Only the first 100 points of the relationship between the envelope kurtosis and the order are plotted in Fig. 11. The largest sudden drop occurs at the second SEK values as can be observed from Fig. 11. Thence, the first two orders are used to reconstruct signal.

The envelope analysis is applied on the above reconstructed signal, and the result is plotted in Fig. 12. The rotational frequency f_r and the fault feature frequency f_o can be picked up and compared with the one in Fig. 10(b); the noise on the pass-band is sparse obviously.

In order to provide evidence of the superiority of the SEK method, the reconstructed order determined by [15] is given

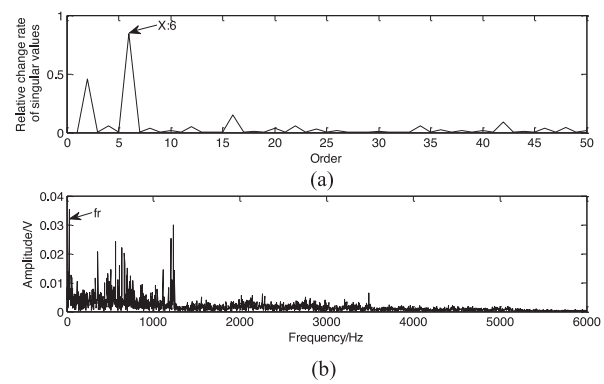

Fig. 13. Analysis result of [15]. (a) Order selected in [15]. (b) Envelope spectrum of [15].

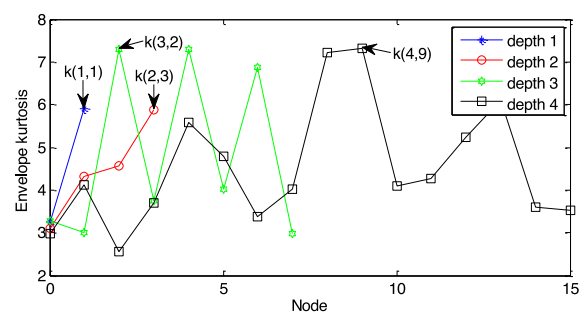

Fig. 14. Envelope kurtosis of different nodes at different depths.

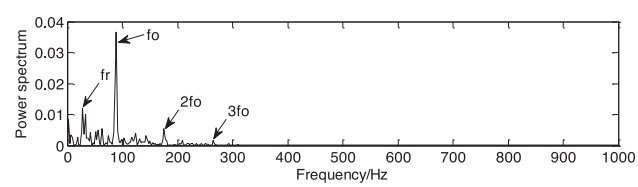

Fig. 15. Envelope spectrum of the optimal node $k^*(4,9)$.

here. The reconstructed order is 6 in Fig. 13(a). The envelope spectrum is plotted in Fig. 13(b). Only the rotational frequency f_r can be extracted from it. Thus, the conclusion that for determining the model order the presented method has obvious advantage can be easily drawn. In addition, δ of Fig. 12 is 1.4, whereas δ of Fig. 13(b) is just 0.57. This also provides evidence for the advantage of SEK.

2) IWPT Analysis: The reconstructed signal is analyzed by IWPT. At each depth, an envelope kurtosis maximum is obtained, respectively: $k(1,1)(5.9114)$, $k(2,3)(5.8874)$, $k(3,2)(7.2951)$, and $k(4,9)(7.3254)$. The global maximum is obtained as $k(4,9)$. Therefore, the optimal node is $k^*(4,9)$. Fig. 14 shows the values of envelope kurtosis for each node.

The selected optimal node $k^*(4,9)$ is reconstructed. The envelope power spectrum is plotted in Fig. 15. The rotational frequency f_r, the fault feature frequency f_o, and its harmonics $(2f_o, 3f_o)$ are very evident. In addition, from the entire frequency axis, there is almost no noise influence on the passband, making the entire frequency axis looks clean.

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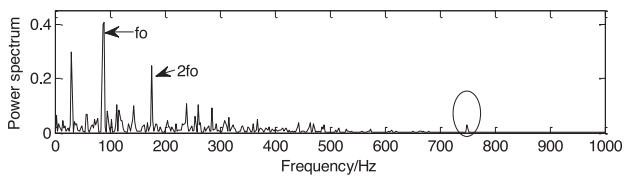

Fig. 16. Analysis result is based on maximum kurtosis.

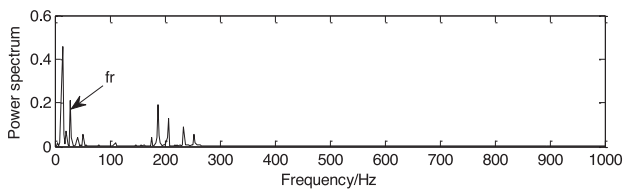

Fig. 17. Analysis result based on minimum entropy.

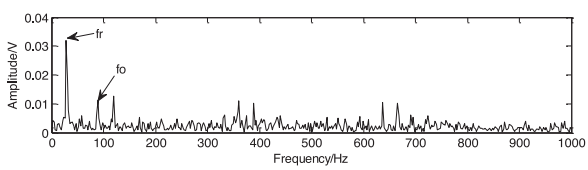

Fig. 18. Envelope spectrum of [14].

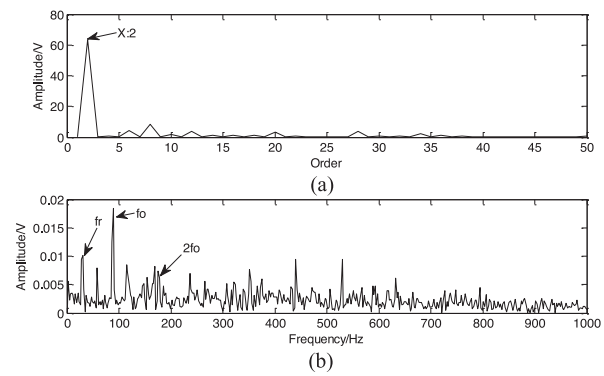

Fig. 19. Analysis results of [19]. (a) Difference spectrum. (b) Envelope spectrum.

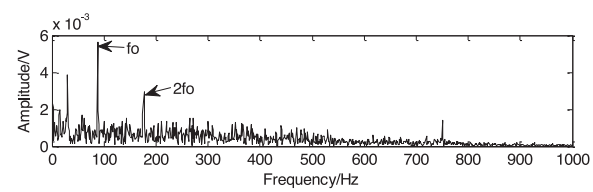

Fig. 20. Analysis results in [23].

Here, the kurtosis maximum is used to select the best node (in [23]) to certify the ascendancy of the SEK method. WPT is applied to the reconstructed signal, and the node is chosen using the kurtosis maximum. After the analysis, the kurtosis value of node $k(4,4)$ is the largest. Similarly, the envelope power spectrum analysis is performed on $k(4,4)$ as shown in Fig. 16. From the figure, although the rotational frequency f_r and the fault feature frequency f_o $(2f_o)$ can be extracted, $3f_o$ cannot be extracted (the proposed method can) and there is a lot of noise interference on the low frequency band (a frequency peak at 750 Hz (inside the black oval frame), especially in the envelope spectrum, which is weakened here). Additionally, indicator δ is used as an auxiliary tool. δ of Fig. 15 is 10.32, whereas δ of Fig. 16 is 6.17. From the above analysis, although the node selected based on the maximum kurtosis can pick up the fault feature frequency well, it is not as good as the CEKM method.

In addition, information entropy is utilized to illustrate the superiority of the CEKM. The entropy value of $k(4,0)$ is minimum and, thus, it is selected as the optimal node. The envelope power spectrum is plotted in Fig. 17, in which the information related to the fault cannot be extracted. Judging by the visibility, the effect of Fig. 17 is obviously not as good as of Fig. 15. In addition, the indicator $\delta = 0.1$ is calculated to compare it with the δ value of Fig. 15. This also shows the advantage of choosing the optimal node based on CEMK.

3) Comparisons With Other Methods: Here, the analysis results of several other methods are given.

First, the method in [14] is applied here. The reconstructed order is 2. From Fig. 18, we can note that some of the passband noise is removed compared to Fig. 10(b), but the overall desiccation capability is weak and interfered by noise. When it is compared with Fig. 15, obviously, the introduced method has a considerable advantage. Additionally, the value of δ is 1.4, which is much smaller than that of E-SVD.

Second, the method in [19] is used. From the difference spectrum of IMF1, plotted in Fig. 19(a), the model order is determined to be 2. Then, the analysis result is plotted in Fig. 19(b), although the rotational frequency f_r, the fault feature frequency f_o, and the harmonic $(2f_o)$ can be extracted. The entire frequency band is disturbed by noise. When it is compared with Fig. 15, the E-SVD has a significant advantage. Similarly, $\delta = 3.3$ is calculated, which is smaller than the value of Fig. 15. Thus, the advantage of E-SVD is also explained.

Third, the method in [23] is utilized here. The kurtosis value of node $k(4,4)(8.4287)$ is the maximum, so $k(4,4)$ is selected as the optimal node for analysis, as shown in Fig. 20. Comparing it with Fig. 15, it can be concluded that the method presented in this article performs better. The indicator $\delta = 5.1$, which is smaller than the value of Fig. 15.

B. Case 2: Actual Bearing Inner-Race Fault Signal

Fatigue tests were conducted and then a series of acceleration data were collected. In order to verify the proposed method, a section of data is taken for analysis. When the initial failure of the bearing occurs, it is determined to be an outer-race failure. The sample frequency is $f_s = 51.2$ KHz. The length of 10 240 sampling is used. Its rotational frequency is $f_r = 50$ Hz, and the fault feature frequency of outer-race is $f_o = 180$ Hz.

The analysis results are provided in Fig. 21(a) and (b), respectively. The effect of extracting the fault feature frequency is not

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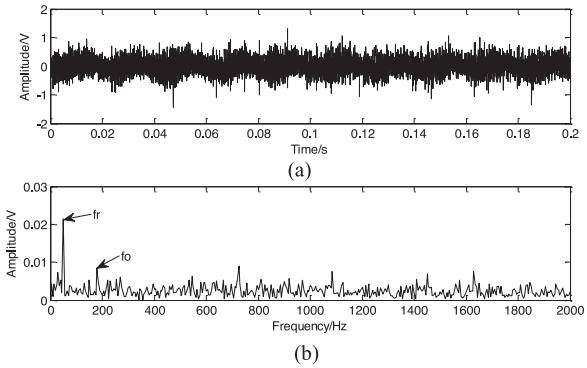

Fig. 21. Original signal. (a) Original signal of outer race fault in time domain. (b) Envelope spectrum.

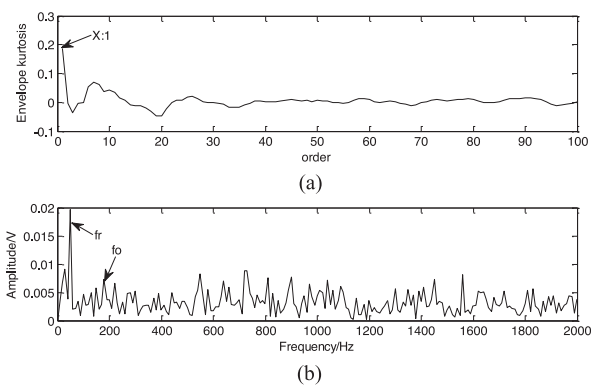

Fig. 22. Result of ISVD. (a) SEK_i. (b) Envelope spectrum.

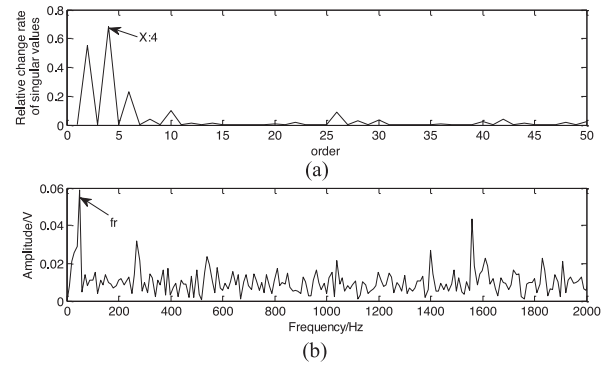

Fig. 23. Analysis result by [15]. (a) Relative-change rate of singular values. (b) Envelope spectrum of by [15].

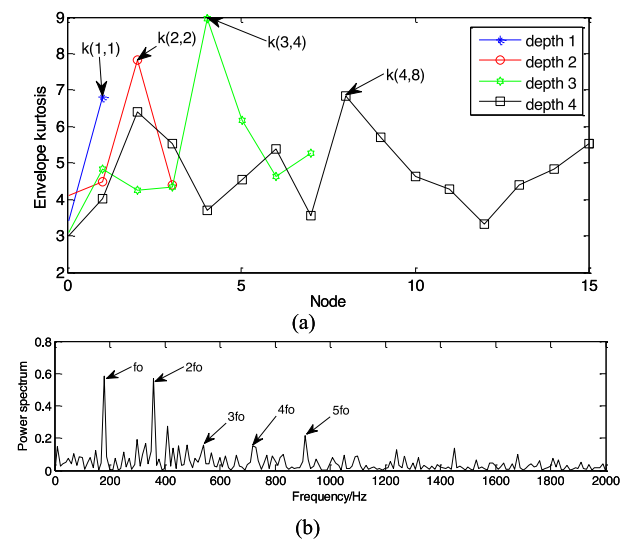

Fig. 24. Result of IWPT analysis. (a) Envelope kurtosis of different nodes at different depths. (b) Envelope spectrum of the optimal node $k^*(3,4)$.

ideal. The introduced E-SVD is utilized to analyze the original signal.

1) ISVD Analysis: The original signal is analyzed by ISVD at first. The results are plotted in Fig. 22. The largest sudden drop occurs at the first SEK value shown in Fig. 22(a). Therefore, the first order is selected for reconstruction. From Fig. 22(b), we can see that the noise over the entire broadband becomes more sparse. However, the noise still strongly interferes with the extraction of fault features. Therefore, further analysis is needed. The indicator $\delta = 1.6$ is calculated.

The model order that is selected by [15] is provided in Fig. 23 to provide the evidence of superiority of the SEK method. The reconstructed order is 4. Thus, the envelope spectrum of the reconstructed signal can be achieved in Fig. 23(b). We can only extract the rotational frequency from it. Therefore, the conclusion can be easily drawn that the presented method for determining the model order has significant ascendancy. The indicator $\delta = 0.89$ is calculated, which is smaller than the value of Fig. 22(b). This also demonstrates that SEK is advantageous.

2) IWPT Analysis: The reconstructed signal is then analyzed by IWPT. Calculating the envelope kurtosis values of all nodes at different depths i ($i = 1, 2, 3, 4$). At each layer, an envelope kurtosis maximum is obtained, respectively: $k(1,1)(6.8122)$, $k(2,2)(7.8250)$, $k(3,4)(8.9636)$, and $k(4,8)(6.8446)$. The global maximum $k(3,4)$ can be obtained. Thus, the optimal node is $k^*(3,4)$. Fig. 24(a) shows the envelope kurtosis values for each node at levels 1, 2, 3, and 4.

The envelope power spectrum of the $k^*(3,4)$ is plotted in Fig. 24(b). The fault signature frequency f_o and the harmonics ($2f_o$, $3f_o$, $4f_o, \ldots$) can be identified clearly. It can be seen that the proposed method can achieve fault diagnosis of the actual bearing effectually. The indicator $\delta = 7.9$ is calculated.

The node selected by the kurtosis maximum is consistent with the node selected by CEKM. Here, the entropy is utilized to illustrate the superiority of the CEKM. In different layers (1–4), the entropy value of $k(4,8)$ is the minimum, thus it is selected as the optimal node. The envelope power spectrum is plotted in Fig. 25, in which the fault signature frequency and the second harmonic can be extracted. Fig. 25 is distinctly not as good as Fig. 24(b). In addition, $\delta = 3.35$, which is much smaller than the value of Fig. 24(b).

3) Comparisons With Other Methods: Here, the E-SVD method is compared with the methods in [14], [19], and [23].

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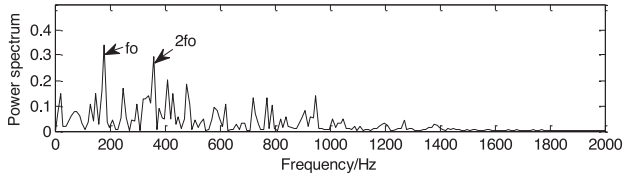

Fig. 25. Analysis result is based on minimum entropy.

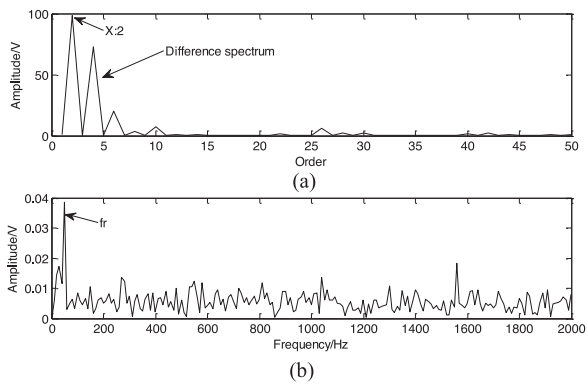

Fig. 26. Analysis result of [14]. (a) Difference spectrum. (b) Envelope spectrum.

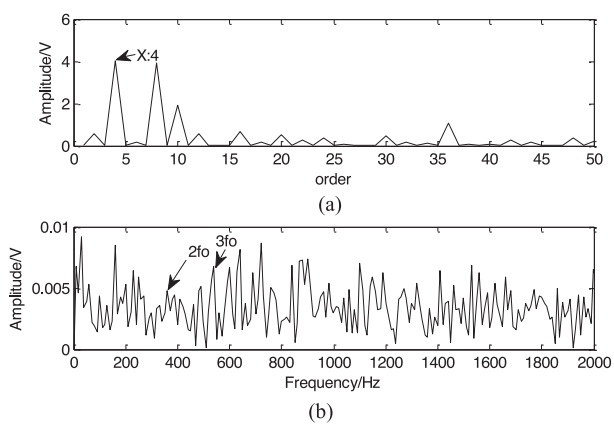

Fig. 27. Analysis results by the method of [19]. (a) Difference spectrum. (b) Envelope spectrum of [19].

First, the method in [14] is utilized here. The reconstructed order is determined to be 2 [see Fig. 26(a)]. In Fig. 26(b), we can note that some of the pass-band noise is removed compared with Fig. 21(b), but the overall recognizable capability is weak. The conclusion can be easily drawn that the E-SVD has considerable advantage, when Fig. 26(b) is compared with Fig. 24(b). $\delta = 1.25$ is calculated, which is much smaller than that of Fig. 24(b).

Then, the method in [19] is applied here. The model order is determined to be 2 [see Fig. 27(a)]. The envelope spectrum is depicted in Fig. 27(b); we can hardly extract the second and third harmonics of the fault characteristic frequency. However, the entire frequency band is disturbed by noise. It is clear that the presented method has a significantly better performance.

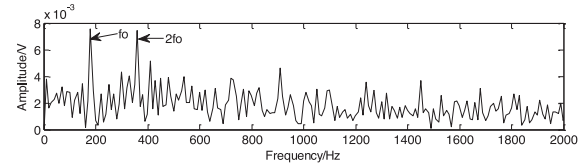

Fig. 28. Analysis result by [23].

TABLE I
EVALUATION INDEX δ OF FOUR METHODS IN CASE 1 AND CASE 2

Cases \ Methods	SVD	EMD-SVD	Improved kurtogram	E-SVD
Case 1	1.4	3.3	5.1	10.32
Case 2	1.25	1.72	5.78	7.9

Similarly, $\delta = 1.72$ is calculated, which is obviously much smaller.

Finally, the method in [23] is applied. Only the analysis result is given. The kurtosis value of node $k(3,4)(6.2400)$ is the maximum; thus, it is selected as the optimal node for analysis, as depicted in Fig. 28. Comparing it with Fig. 24(b), obviously, the presented method has significant advantages. Similarly, $\delta = 5.78$ is calculated to compare with δ of Fig. 24(b), it is smaller. Therefore, the conclusion can be drawn that the introduced E-SVD method is more advantageous than this method in this case.

In addition to direct analysis and comparison from the spectrums, the quantitative index δ is introduced and used as an auxiliary analysis tool, as described in Case1 and Case2. In order to give a more visual comparison of the four methods (based on indicator δ), the comparison of the four methods of indicator δ in Case 1 and Case 2 is given below in Table I. Obviously, the δ value of the result of the E-SVD method proposed in this article is significantly larger than the other three methods, whether in Case 1 or Case 2. This also provides supporting evidence of the superiority of the E-SVD method in this article.

In conclusion, in this section, the E-SVD method is smoothly applied to the actual cases. Regardless of Case 1 or Case 2, it has more robust performance than the other three commonly used methods. Naturally, the conclusion that the proposed E-SVD method has considerable advantages can be easily drawn.

V. CONCLUSION

In this article, E-SVD was presented as an innovative method to achieve the fault diagnosis of rotating machinery. First, a novel indicator SEK was introduced and first used to determine the valid reconstructed order of SVD. SEK was applied to the analysis of simulated signal and actual cases with satisfactory results successfully. Thus, the innovative method ISVD was constructed based on SEK. Second, a novel node selection method based on CEKM was presented to select the optimal node of WPT, which gives birth to IWPT. The CEKM was excellent and stable than kurtosis and entropy. The IWPT was used to enhance the noise reduction capability of ISVD. Therefore, the proposed E-SVD was an organic combination of ISVD and IWPT, combining ISVD's pass-band random noise cancellation capability and

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IWPT's excellent bandpass filter design capability. It took full advantages of the two methods, which greatly improved the noise reduction capability, and was corroborated by examples in the text. Finally, an indicator was first introduced to characterize the E-SVD effect. The early fault characteristic frequency of the rolling bearing was effectively extracted by E-SVD to realize the fault diagnosis. It was compared with some typical methods, and all the results indicated that the E-SVD has more excellent performance. This can be the evidence that the E-SVD has strong application potential in rotating machinery. We can look forward to it being recognized in the future; the SEK-based reconstructed order determined method of SVD, the CEKM-based optimal node selected method of WPT, and the fault diagnosis method of E-SVD introduced in this article can be effectively applied to rotating machinery and even other fields.

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